PROJECT DOCUMENT

Mitsue Project

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The Mitsue Project

Ecology · Energy · Digital Infrastructure for Rural Japan

> **"The forest our ancestors planted — powering the village they built."** 先人が植えた森が、今、村に豊かさをもたらす。

A 25-year initiative to revitalise the former Sugano Elementary School and the forest that surrounds it

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What the Project Is

The Mitsue Project transforms the **former Sugano Elementary School (Mitsue Taiken Koryukan)** — a community exchange center available for reuse, the leading candidate site (final choice confirmed in Phase 1) — and the aging cedar forest above it, into a living model of rural self-sufficiency. Three elements work together:

- **Forest restoration** — Replacing aging sugi monoculture with native species, healing land ecologically depleted for decades. Thinning the sugi also yields the biomass fuel that powers the village. Native broadleaf trees feed deer, wild boar, and bear — keeping wildlife in the forest and out of crop areas
- **Biomass CHP energy** — Sugi forest thinnings fuel a combined heat and power (CHP) system generating electricity (primary) and heat (secondary), complemented by solar panels and EV charging for residents and visitors, with battery backup during grid outages. A circular local energy economy where the forest powers the village. Battery storage to be evaluated in feasibility study
- **Sustainable AI data center** — A small, efficient digital facility housed in the former Sugano Elementary School (Koryukan) or a disused village factory building — the leading candidate is the former Sugano school; final site confirmed in Phase 1 — powered entirely by local energy, providing infrastructure and employment where none existed. A working model of progress and sustainability as a single coherent system

These are not three separate ideas. Each one makes the others possible.

The Problems This Project Solves

Challenge	How the Project Responds
Former Sugano school and disused village factories vacant, underused	Leading candidate (former Sugano school) or alternative (disused factory) repurposed as data center & community anchor — site confirmed Phase 1
Cedar monoculture — ecologically depleted	Systematic conversion to native mixed forest
Energy system needs reliable round-the-clock demand	Data center provides 24/7 baseload demand that justifies the biomass CHP and solar investment
Rural grid dependency, no local power	Biomass CHP from sugi thinnings (electricity + heat) as the primary supply, complemented by solar, EV charging, and battery storage
Shrinking population, no local jobs	Data center operations, forestry, maintenance roles
Blackouts threaten critical services	On-site biomass CHP generation provides 24/7 baseload power, keeping the Koryukan, data center, and key facilities running during outages

Challenge	How the Project Responds
Wildlife raiding crop areas	Restored native broadleaf forest provides natural food — acorns, nuts — keeping deer, wild boar, and bear in the forest
No local EV charging for rural residents	Mitsue builds EV-ready charging infrastructure for residents and visitors; the local biomass-and-solar energy system powers it sustainably, independent of the national grid
Existing rural blackouts	Aging distribution lines already cause outages in Mitsue; a village-owned power supply provides resilience and keeps critical services running

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Who Is Behind It

Rob Oudendijk — Dutch electrical engineer, resident of Mitsue since 2012, core hardware developer for Safecast, the global open citizen science network born after Fukushima.

Advisors: Ray Ozzie (Executive Chair, Blues) · San Poisson (Project Manager) · Takuo Dome (Professor Emeritus, Osaka University) · Evin Zoet (Co-Representative Director, Transom) · Yoshiko Zoet-Suzuki (Co-Representative Director, Transom)

A dedicated **non-profit** is being established with local residents, village leadership, forestry professionals, and academic partners.

Why It Can Work Here

Mitsue has what most villages lack: the former Sugano Elementary School (Koryukan) available for reuse, productive cedar forest within reach, a river for water cooling, and a resident with both the technical background and the deep local relationships to bridge these worlds.

The Mitsue Project — What to Expect, and When

Benefits Arrive Well Before Year 25

The 25-year horizon describes full forest restoration — not how long the village waits for results.

Timeline	Milestone
Year 1	Former Sugano school (Koryukan) reactivated as datacenter hub; international media visibility for Mitsue
Year 2	Landowners begin receiving income from cedar harvest
Year 3–4	EV charging stations operational; battery storage if confirmed by feasibility study

Timeline	Milestone
Year 4–5	Data center operational; local employment in operations and maintenance
Year 5–10	Data service revenue reinvested into village and forest programmes
Year 25	Native forest established; model documented for replication worldwide

How It Is Financed

The project stacks complementary sources rather than depending on a single funder: **林野庁 grants** (forest restoration) · **METI / NEDO grants** (rural energy resilience and EV infrastructure) · the village-led **地域脱炭素移行・再エネ推進交付金** (2/3–3/4 subsidy on solar/battery/EV, via the village's official 2025 renewable-energy plan) · **FIT / FIP** feed-in tariff revenues · **J-Credit** carbon offset income · **data center service revenue** (primary long-term income) · **impact investment and philanthropy** (Phase 0 and early infrastructure).

All environmental, energy, and forestry data will be published openly — so any village facing the same challenges can learn from what is built here. This ethos comes directly from Safecast, which has published over 250 million measurements openly since 2011.

What Makes This Different

Most rural revitalisation projects in Japan either depend permanently on outside subsidy or import a use with no connection to the land. The Mitsue Project is built around the village's own resources — its forest, its community facilities, its people — and is designed to become financially self-sustaining within the first decade.

The data center element is not incidental. It creates the stable, round-the-clock energy demand that makes the village's biomass CHP and solar generation economically viable — and the on-site biomass baseload power extends naturally to the whole community during blackouts. The three pillars reinforce each other in a way that cannot easily be replicated with only one or two of them. Together they set a working example of how rural communities can integrate technological progress with ecological sustainability — not as opposites, but as a single coherent system.

Current Status

The project is in **Phase 0** — community consultation, feasibility study, and non-profit formation. No major commitments have been made. This is the moment to ask questions, raise concerns, and shape what comes next.

How to Get Involved

There are several ways to support or participate at this early stage:

- **Village residents & landowners** — Join the consultation process; your land, knowledge, and voice shape the design
- **Forestry & energy professionals** — Feasibility input, technical review, and partnership on implementation
- **Investors & philanthropists** — Phase 0 funding for feasibility studies, legal structure, and early outreach
- **Researchers & academics** — Collaborate on open data, carbon measurement, and rural energy modelling

- **Other rural communities** — Follow the project; everything built here will be documented and shared openly

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> *"If it works here, it can work elsewhere."*

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